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Milestone 3
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CS 490
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Milestone 3: Fundamental Analysis GPT

1. Problem Statement and Task Definition

The objective of this project is to build and evaluate an end-to-end Generative Pretrained Transformer (GPT) tailored for the financial domain. The task is split into two phases: (1) pretraining a custom GPT model from scratch to understand the linguistic structure and vocabulary of corporate financial documents, and (2) applying parameter-efficient fine-tuning (PEFT) to adapt the base model into an instruction-following assistant capable of performing financial question answering.

2. Corpus Description

- **Phase 1 (Pretraining):** The base model was trained on the eloukas/edgar-corpus dataset, which consists of U.S. Securities and Exchange Commission (SEC) 10-K and 10-Q filings. This corpus was selected to teach the model corporate jargon, financial boilerplate, and the structural formatting of official market disclosures.
- **Phase 2 (Adaptation):** For fine-tuning, the wandb/finqa-data-processed dataset was utilized. This dataset contains specific financial questions paired with context. The data was reformatted into an "Instruction / Input / Response" string structure to teach the model how to answer specific financial queries rather than merely predicting the next token in a corporate document.

3. Ethics, Privacy, Bias, and Licensing Considerations

- **Privacy & Licensing:** The SEC EDGAR corpus consists of public domain government filings, mitigating major privacy and copyright concerns. However, generating synthetic financial advice or analysis carries significant real-world risk if users rely on the model's outputs for actual investment decisions.
- **Bias:** Training a language model exclusively on corporate filings inherently biases the model's worldview. The outputs are heavily skewed toward corporate jargon ("boilerplate"), risk-averse legal phrasing, and a strictly quantitative perspective, potentially ignoring the qualitative human impacts of corporate actions.

4. Method Overview

The project pipeline consists of a custom generative pretraining workflow followed by an adaptation phase. Data was streamed and tokenized using the tiktoken library (gpt2 encoding) to ensure vocabulary consistency across both phases. The architecture is a custom-built, Pre-LayerNorm GPT. Following the pretraining phase, the model was adapted using Low-Rank Adaptation (LoRA) to enable instruction-following without updating the entire parameter space.

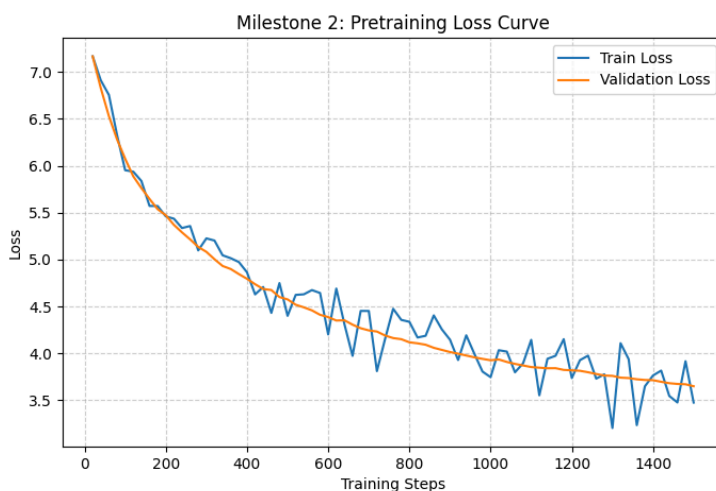
5. Model and Training/Adaptation Setup

- **Base Model Setup:** The architecture is a custom GPT model featuring 6 transformer layers, 8 attention heads, a 512 embedding dimension, and a 256-token context length. Training was executed on an NVIDIA A100 GPU and accelerated using PyTorch 2.0's torch.compile graph compiler.
- **LoRA Adaptation Setup:** Parameter-efficient fine-tuning was achieved by injecting LoRA adapters into the Attention Block's Query (W_q) and Value (W_v) projection matrices. The LoRA hyperparameters were set to a Rank of 8 and an Alpha of 16. Out of the ~70,607,872 total model parameters, only 98,304 parameters were trainable during this phase, successfully freezing 99.86% of the network.

6. Evaluation Setup

The comparison between the base pre-trained system and the LoRA-adapted system was conducted using a strict, fair evaluation protocol. Both models were evaluated using the exact same held-out validation set, the identical gpt2 tokenizer, and the same 256 context length. Text generation was tested using identical multinomial sampling parameters (temperature=0.8, top_k=40). The quantitative metrics used for comparison were Validation Loss and Perplexity.

7. Results



Metric	Base Model (Pre-trained)	Adapted Model (LoRA)
Train Loss	3.5442	5.0432
Val Loss	3.6382	4.8341
Val Perplexity	38.02	125.73

Analysis:

The quantitative results demonstrate the performance characteristics of the model across its two distinct training phases. The Base Model achieved a strong validation loss of 3.6382 and a perplexity of 38.02, indicating that it successfully modeled the statistical distribution and learned the underlying grammar of the SEC EDGAR pretraining corpus.

Following the Parameter-Efficient Fine-Tuning (PEFT) process on the FinQA objective, the Adapted Model's validation loss decreased steadily over its 500 update steps, settling at 4.8341 with a validation perplexity of 125.73. The increase in both loss and perplexity between the Base and Adapted models is expected; it reflects the significant cognitive shift from a simple document-continuation task to a much more complex instruction-following task. While the LoRA validation loss tracked slightly lower than its training loss (suggesting the model generalized well to the held-out instruction set without overfitting), the elevated perplexity of 125.73 indicates that the model retains a high degree of uncertainty when predicting exact tokens in this new format. This quantitative uncertainty aligns perfectly with the qualitative failure cases observed in Section 8, where the model struggled to completely unlearn its strong pretraining prior (the SEC boilerplate) and fully commit to the strict Question/Answer extraction format.

8. Error Analysis and Limitations

To evaluate the qualitative performance of the adapted model, a set of financial prompts was passed through the network. The varying outputs highlight the tension between the model's pretraining prior and the fine-tuning objective.

Prompt	Base Model (Complete Output)	Adapted (LoRA) Model (Complete Output)
Item 1A. Risk Factors. As a wholesale power trading entity, our financial condition is highly dependent on grid stability and commodity price fluctuations. During periods of extreme weather, unexpected margin calls from our clearing broker could result in	reduced the Company's ability to generate revenue from its operating results in an effort to reduce costs at attractive prices. The Company seeks to enhance its sales and marketing programs with significant capital expenditures under its own control strategy. The Company believes it can attract high levels of financial resources that are critical to the Company's business. In addition, as well as the consolidation of new systems, the Company faces in many key projects, including those in the development stage. For example, the Company has enhanced capabilities for all	a loss of adverse tax or other regulatory changes in the future to its stock or other investors, which will adversely affect results of operations and financial position. COMPETITION In addition to a variety of factors, including the relative size of its business strategy and the use of new market opportunities that may materially reduce rates on favorable terms and the market prices for customers to the Company's and international trading businesses are also affected by changes in industry guidelines and by some risk conditions such as price and foreign exchange
Management's Discussion and Analysis: Regarding the implied volatility skew observed in our equity derivatives portfolio, particularly the out-of-the-money NVDA call options, the primary driver of this pricing anomaly is	available upon its continued ability to provide a comprehensive range of high quality long distance services in order to improve the efficiency and reduce costs. The Company believes that these efforts are well developed for new market entrants on acceptable terms. The Company's international markets may continue to enter into other markets with local exchange carriers, including those who purchase local carrier carriers' facilities, resellers and their operations. Such foreign markets also generally require private companies to maintain network infrastructure for long distance carriers to offer intrastate rates within	used to collect higher margin and provide for a return on the amount available to customers within two years beginning after 2002. The Company also has an option to establish a base rate structure on its investments before taking into account purchases; however, that the new line will not place as a percentage of the Company's total assets are invested in businesses .', 'net performance is currently a gain or income in the Company's consolidated financial statements was approximately \$ 3 million , compared with \$ 2 million and \$ 542 million

Extract the core financial risk from the following paragraph: Paragraph: The ongoing litigation regarding the delayed deployment of our high-performance computing clusters has severely restricted our operating cash flow, forcing us to draw \$50 million from our revolving credit facility. Core Risk:	Inktomi intends to seek to acquire additional capital stock through a merger agreement (the "Merger Agreement") with First Northern Telecom Corporation ("Brentfield"), which is expected to be merged into an acquisition and merger ("MaineWebber Group"). As a result, the Company will receive two preferred shares, Common Stock that, if at all, upon their transaction, including issuance of certain warrants, options or other obligations to merge or otherwise dispose of its outstanding common stock, and all of its subsidiaries	Total time and costs were included in "expect -- Acquisitions--the Company's financial statements included as discussed herein. There can be no assurance that tARRATIVE DESCRIPTION OF BUSINESS Business: Operations PRIVITY Maturities (1) Innotance, we held various business activities (in which it owns or operates a portfolio (5%) are subject to extensive regulation by the Board of Directors of the Board of Directors upon
On March 15, the Board of Directors announced the immediate resignation of the Chief Financial Officer. To ensure continuity in our algorithmic trading division and to oversee the upcoming quarterly audit, the Board has appointed	a plan that provides for certain other things, such plan and will not raise any capital gain or a vote on the merger with the Company. On February 21, 1997, the Board declared the Board of Directors authorized the Board of Directors of the Company to become President and Chief Executive Officer from January 1, 1999 until December 31, 1997, pursuant to which the Company is responsible as a partner's stockholder, was employed by Mr. John Wuhe held various positions at the time of management and	certain of the Company's operations as part of its accounting for future financial statements and reported information as a result of such forward-looking statements are subject to periodic reports relating to those anticipated or that are not significant to the following business segments: Market The Registrant manufactures high-quality products including a variety of consumer products. These processes are sold primarily through a variety of regional trade showers and independent distributors, including local suppliers and wholesalers, and by mail catalogs as well as department

Based on qualitative testing of the generated outputs, the system demonstrated specific strengths and weaknesses:

- **Success Case (Style Mastery):** When prompted with "Item 1A. Risk Factors," the adapted model successfully generated highly accurate structural boilerplate (e.g., smoothly transitioning into sentences about "adverse tax or other regulatory changes"). It mastered the tone of SEC filings.
- **Failure Case 1 (Dataset Artifacts):** In one generation, the model output literal Python string formatting (... invested in businesses .', 'net performance is currently a gain...). This indicates that the raw huggingface dataset contained uncleaned arrays or code artifacts, which the model memorized.
- **Failure Case 2 (The Pretraining Prior):** When given a strict FinQA extraction instruction, the model failed to answer the prompt and instead continued generating random corporate boilerplate.
- **Failure Case 3 (Context Degradation / Hallucination):** When prompted with the scenario of a CFO resigning and the Board appointing a successor, the model began its generation coherently but rapidly lost context. Its output: "...the Board has appointed certain of the Company's operations as part of its accounting for future financial statements..." before completely devolving into irrelevant boilerplate about manufacturing consumer products and selling through mail catalogs. This is a classic example of context degradation, where the model loses the semantic thread of the prompt and falls back on high-frequency, generic pretraining tokens that have no logical connection to the specific scenario.
- **Limitation:** The failure to follow instructions perfectly indicates that 500 LoRA update steps with a Rank of 8 was not enough training volume to force the model to unlearn its base pretraining behavior and switch to a strict Question/Answer format.

9. Lessons Learned

- Building the PyTorch architecture, with the attention mechanisms, the layer norms, and the feed-forward networks, is ultimately just constructing an empty vessel. The fundamental behavior, tone, and capability of the model are entirely dictated by the pretraining data distribution. Training strictly on the SEC EDGAR corpus resulted in a model that possessed a command of corporate legal syntax and financial boilerplate, demonstrating that an LLM does not learn "facts," but rather the statistical worldview of its training set.
- The transition from the pretraining phase to the FinQA fine-tuning phase highlighted the core challenge of modern AI: bridging the gap between statistical mimicry and logical instruction-following. The base model excelled at "document continuation," predicting the next logical corporate phrase. However, forcing the model to stop continuing a document and instead extract a specific answer to a prompt proved highly resistant. It

revealed that achieving true zero-shot reasoning requires a massive volume of instruction-tuning to override the model's deeply ingrained probabilistic prior.

- Implementing Low-Rank Adaptation (LoRA) provided a profound lesson in the mechanics of neural representations. By freezing 99.86% of the network and only updating the query and value projection matrices, the model's behavior was successfully nudged toward a Question/Answer format. This demonstrated that a model's core linguistic understanding (the frozen weights) is entirely separable from its interaction format (the LoRA adapters). Understanding this decoupled nature of LLMs is essential, as it proves that massive, expensive models can be highly specialized for niche financial tasks without requiring a full recalculation of the baseline parameter space.